1-Number of Zeros in a Given Array

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer:(penalty regime: 0 %)

```c
#include <stdio.h>

int firstZero(int arr[], int low, int high, int n) {
    if (low > high)
        return n;

    int mid = (low + high) / 2;

    if (arr[mid] == 0 && (mid == 0 || arr[mid - 1] == 1))
        return mid;
    else if (arr[mid] == 1)
        return firstZero(arr, mid + 1, high, n);
    else
        return firstZero(arr, low, mid - 1, n);
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int index = firstZero(arr, 0, n - 1, n);
    printf("%d\n", n - index);

    return 0;
```

}

Feedback

	Input	Expected	Got	
	5 1 1 1 0 0	2	2	
	10 1 1 1 1 1 1 1 1 1 1	0	0	
	8 0 0 0 0	8	8	

	Input	Expected	Got	
	0 0 0 0			
	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	

Passed all tests!